DM 1ST.py - C:/Users/thala/Downloads/FODS/DM 1ST.py (3.14.3)

File Edit Format Run Options Window Help

```python
import pandas as pd
import numpy as np

# Sample data with missing entries
data = {
    'Name': ['Alice', 'Bob', 'Charlie', 'David'],
    'Age': [25, np.nan, 30, 22],
    'Salary': [50000, 60000, np.nan, 45000]
}

df = pd.DataFrame(data)

# 1. Check for missing values
print("Missing count:\n", df.isnull().sum())

# 2. Fill missing Age with the column median
df['Age'] = df['Age'].fillna(df['Age'].median())

# 3. Drop rows where Salary is missing
df = df.dropna(subset=['Salary'])

print("\nCleaned DataFrame:\n", df)
```

IDLE Shell 3.14.3

File Edit Shell Debug Options Window Help

```
Python 3.14.3 (tags/v3.14.3:323c59a, Feb  3 2026, 16:04:56) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
=============== RESTART: C:/Users/thala/Downloads/FODS/DM 1ST.py ===============
Missing count:
 Name      0
Age       1
Salary    1
dtype: int64

Cleaned DataFrame:
     Name   Age   Salary
0  Alice  25.0  50000.0
1    Bob  25.0  60000.0
3  David  22.0  45000.0
>>>
```

Ln: 16 Col: 0

DM 2ND.py - C:/Users/thala/Downloads/FODS/DM 2ND.py (3.14.3)

File Edit Format Run Options Window Help

```python
import pandas as pd

data = {
    'Employee': ['Amy', 'John', 'Raj', 'Kevin', 'Sora'],
    'Department': ['HR', 'IT', 'IT', 'Sales', 'IT'],
    'Experience': [3, 7, 1, 5, 6]
}

df = pd.DataFrame(data)

# Filter for employees in IT with more than 5 years of experience
filtered_df = df[(df['Department'] == 'IT') & (df['Experience'] > 5)]

print(filtered_df)
```

IDLE Shell 3.14.3

File Edit Shell Debug Options Window Help

```
Python 3.14.3 (tags/v3.14.3:323c59a, Feb  3 2026, 16:04:56) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
============== RESTART: C:/Users/thala/Downloads/FODS/DM 1ST.py ==============
Missing count:
 Name      0
Age       1
Salary    1
dtype: int64

Cleaned DataFrame:
     Name   Age   Salary
0  Alice  25.0  50000.0
1    Bob  25.0  60000.0
3  David  22.0  45000.0
>>>
============== RESTART: C:/Users/thala/Downloads/FODS/DM 2ND.py ==============
  Employee Department  Experience
1     John         IT           7
4     Sora         IT           6
>>>
```

Ln: 21 Col: 0

DM 3RD.py - C:/Users/thala/Downloads/FODS/DM 3RD.py (3.14.3)

File Edit Format Run Options Window Help

```python
import pandas as pd

data = {
    'Store': ['North', 'South', 'North', 'West', 'South', 'West'],
    'Sales': [1200, 1500, 900, 1100, 1700, 1300],
    'Employees': [5, 7, 5, 4, 8, 4]
}

df = pd.DataFrame(data)

# Group by Store and calculate total sales and average employees
summary = df.groupby('Store').agg({
    'Sales': 'sum',
    'Employees': 'mean'
}).reset_index()

print(summary)
```

IDLE Shell 3.14.3

File Edit Shell Debug Options Window Help

```
Python 3.14.3 (tags/v3.14.3:323c59a, Feb  3 2026, 16:04:56) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
=============== RESTART: C:/Users/thala/Downloads/FODS/DM 1ST.py ===============
Missing count:
 Name      0
Age       1
Salary    1
dtype: int64

Cleaned DataFrame:
     Name   Age   Salary
0  Alice  25.0  50000.0
1    Bob  25.0  60000.0
3  David  22.0  45000.0
>>>
=============== RESTART: C:/Users/thala/Downloads/FODS/DM 2ND.py ===============
  Employee Department  Experience
1     John         IT           7
4     Sora         IT           6
>>>
=============== RESTART: C:/Users/thala/Downloads/FODS/DM 3RD.py ===============
   Store  Sales  Employees
0  North   2100        5.0
1  South   3200        7.5
2   West   2400        4.0
>>>
```

Ln: 27 Col: 0

DM 4TH.py - C:/Users/thala/Downloads/FODS/DM 4TH.py (3.14.3)

File Edit Format Run Options Window Help

```python
import pandas as pd

data = {
    'Age': [25, 30, 35, 40, 45],
    'Salary': [50000, 60000, 75000, 90000, 110000],
    'Department': ['HR', 'IT', 'IT', 'Marketing', 'HR']
}

df = pd.DataFrame(data)

print("--- Numeric Summary ---")
print(df.describe())

print("\n--- Categorical Summary ---")
print(df.describe(include='str'))
```

IDLE Shell 3.14.3

File Edit Shell Debug Options Window Help

```
Python 3.14.3 (tags/v3.14.3:323c59a, Feb  3 2026, 16:04:56) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
=============== RESTART: C:/Users/thala/Downloads/FODS/DM 4TH.py ===============
--- Numeric Summary ---
             Age         Salary
count   5.000000       5.000000
mean   35.000000   77000.000000
std     7.905694   23874.672773
min    25.000000   50000.000000
25%    30.000000   60000.000000
50%    35.000000   75000.000000
75%    40.000000   90000.000000
max    45.000000  110000.000000

--- Categorical Summary ---
       Department
count           5
unique          3
top            HR
freq            2
>>>
```

Ln: 22 Col: 0

DM 4TH.py - C:/Users/thala/Downloads/FODS/DM 4TH.py (3.14.3)

DM 5TH.py - C:/Users/thala/Downloads/FODS/DM 5TH.py (3.14.3)

File Edit Format Run Options Window Help

```python
import pandas as pd

# Creating a sample dataset
data = {
    'Name': ['Alice', 'Bob', 'Charlie', 'David', 'Eva', 'Frank
    'Score': [85, 92, 78, 95, 88, 72, 91]
}

df = pd.DataFrame(data)

# 1. Get the top 5 rows (default behavior)
print(df.head())

# 2. Get the top 2 rows
print(df.head(2))

# 3. Get the bottom 3 rows
print(df.tail(3))
```

IDLE Shell 3.14.3

File Edit Shell Debug Options Window Help

```
Python 3.14.3 (tags/v3.14.3:323c59a, Feb  3 2026, 16:04:56) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
=============== RESTART: C:/Users/thala/Downloads/FODS/DM 4TH.py ===============
--- Numeric Summary ---
             Age         Salary
count   5.000000       5.000000
mean   35.000000   77000.000000
std     7.905694   23874.672773
min    25.000000   50000.000000
25%    30.000000   60000.000000
50%    35.000000   75000.000000
75%    40.000000   90000.000000
max    45.000000  110000.000000

--- Categorical Summary ---
       Department
count           5
unique          3
top            HR
freq            2
>>>
=============== RESTART: C:/Users/thala/Downloads/FODS/DM 5TH.py ===============
      Name  Score
0    Alice     85
1      Bob     92
2  Charlie     78
3    David     95
4      Eva     88
    Name  Score
0  Alice     85
1    Bob     92
    Name  Score
4    Eva     88
5  Frank     72
6  Grace     91
>>>
```

Ln: 37 Col: 0